THE PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
NAMANGA SECONDARY SCHOOL
FORM FOUR PRE-MOCK EXAMINATION
033 BIOLOGY
MARKING SCHEME APRIL 2026

SECTION A (16 MARKS)

1.

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
C	B	C	E	A	E	A	A	C	C

01@ = 10 marks

2.

LIST A	(i)	(ii)	(iii)	(iv)	(v)	(vi)
LIST B	I	G	F	D	C	B

01@ = 06 marks

SECTION B (54 MARKS)

3. (a)
   (i) A special room or building designed and equipped for performing biological experiments, research, and observations. *01½ marks*
   (ii) Specimen is an actual sample of an organism (e.g., a plant, animal, or part like a tissue) used for direct examination while a model is an object for physical representation of biological structure. *03 marks*

   (b) Importance of Lab Rules:
   (i) No eating/drinking: To prevent accidental ingestion of toxic chemicals or pathogens.
   (ii) Lab coat/goggles: To protect the skin and eyes from chemical splashes and heat.
   (iii)Labeling containers: To ensure the correct chemicals are used and to prevent dangerous mixing of unknown substances. *01½ @ = 04½ marks*

(Total: 09 marks)

4. (a)
   (i) At trophic levels, species are categorized as autotrophs (or primary producers) example, green plants (1½ marks), *heterotrophs (or consumers)*, and Detritivores (or decomposers), examples bacteria and fungi. ((1½ marks)
   (ii) Heterotrophs can be further sub-divided into different functional groups, including primary consumers (herbivores), secondary consumers (carnivorous predators that feed exclusively on herbivores) and tertiary consumers (predators that feed on a mix of herbivores and predators). (03 marks)

   (b) Because predators are higher up the food chain. In the food chain an organism passes only part of the energy it receives from food. With less energy, each level in the food chain supports fewer individuals than the one below. (03 marks)

(Total: 09 marks)

5. (a) Diagram: *(drawing should show: Left/Right Kidneys, Ureters leading to the Urinary Bladder, and the Urethra. Label the Renal Artery and Renal Vein)*

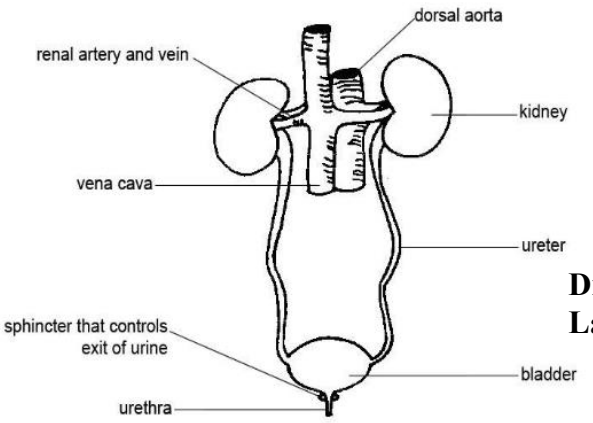


Diagram: 02 ½ pts
Labels Any 4 × 0½ =02

(b) Functions:

(i)

- Osmoregulation: Controlling the water and salt balance in the blood.
- Excretion**:** Removing metabolic waste products like urea and uric acid from the blood. ***03 marks***

(ii) During cold there is no sweating, less ADH hormone will be produced that will cause less reabsorption of water, hence more water will be excreted as urine.

01½ marks **(Total: 09 marks)**

6. (a) Blood groups cannot be used to prove paternity, However, they can show that the male is not a biological father of a particular child, as a child cannot have a blod group antigen that is not present in either parent, i.e, if a child is group A, a father is group O, and a mother ia group O, the man cannot be the biological father of the child. ***(03 marks)***

(b)

(i) There are three alleles that control the ABO blood group: I^A, I^B. and i. The allele I^A is responsible for the formation of Antigen A for blood group A, The allele I^B is responsible for the formation of Antigen B for blood group B, and the allele i is responsible for no Antigen formation. Both I^A, and I^B are dominant so if I^A and I^B are present together they form blood group AB and i is recessive to both I^A and I^B .

so if you have the genotype I^A i you will have blood group A and if you have the genotype I^B i you will have blood group B. However, if you have the genotype ii then you are homozygous for i and will be of blood group O.

Phenotype (Blood group)	Genotype
A	I^AI^A or I^Ai
B	I^BI^B or I^Bi
AB	I^AI^B
O	ii

(01½ mark)

(ii)

	Phenotype (Blood group)	Genotype
Husband	A	I^AI^A or I^Ai
Wife	B	I^BI^B or I^Bi

It is possible for the man to be the biological parent of the child with blood group O if

	Phenotype (Blood group)	Genotype
Husband	A	I^Ai
Wife	B	I^Bi

(01½ mark)

Husband X Wife

		Husband	
		I^A	i
Wife	I^B	$I^B I^A$	I^B i
	i	I^A i	i i

(01mark)

(iii)

Husband X Wife

		Husband	
		I^A	i
Wife	I^B	$I^B I^A$	I^B i
	i	I^A i	i i

1 group AB: 1 group A: 1 group B: 1 group O

or

Husband X Wife

		Husband	
		I^A	I^A
Wife	I^B	$I^A I^B$	$I^A I^B$
	i	I^A i	I^A i

2 group AB: 2 group A:

Or Husband X Wife

		Husband	
		I^A	i
Wife	I^B	$I^A I^B$	I^B i
	I^B	$I^A I^B$	I^B i

2 group AB: 2 group B:

Or Husband X Wife

		Husband	
		I^A	I^A
Wife	I^B	$I^A I^B$	$I^A I^B$
	I^B	$I^A I^B$	$I^A I^B$

All group AB ***((00½) @ = 2 marks)*** **(Total: 09 marks)**

7. (a) Transpiration: The process by which plants lose water vapor through the stomata of their leaves to the atmosphere. ***01½ marks***

   (b) Effects:

   (i) Temperature

   Effect: Increases transpiration rate.

   Mechanism: Higher temperatures increase the **kinetic energy** of water molecules, allowing them to move faster and evaporate more quickly from the mesophyll cell surfaces into the air spaces of the leaf. Additionally, warmer air has a higher capacity to hold water vapor, which lowers its relative humidity and maintains a steep diffusion gradient. ***02½ marks***

   (ii) Humidity

   Effect: Decreases transpiration rate (High humidity decreases the rate as the air is already saturated.). ***02½ marks***

   Mechanism: Transpiration is inversely proportional to relative humidity. High humidity means the air already contains a high concentration of water vapor. This reduces the **diffusion gradient** between the saturated air inside the leaf and the atmosphere outside, making it harder for more water vapor to escape.

   (ii) Wind Speed

   Effect: Increases transpiration rate. ***02½ marks***

   Mechanism: In still air, water vapor accumulates around the leaf surface, forming a "boundary layer" of humid air that slows down further evaporation. Wind sweeps away this saturated air and replaces it with drier air. This maintains a high concentration gradient for water vapor to diffuse out of the stomata.

(Total: 09 marks)

8. (a)

Rib cage

   (i) **Protection of Vital Organs:** Its primary role is to shield delicate thoracic organs, specifically the **heart** and **lungs**, as well as major blood vessels from external trauma.

   (ii) **Mechanical Aid in Respiration:** The ribs are essential for breathing. By moving upward and outward during inhalation, they increase the volume of the thoracic cavity, lowering internal pressure to draw air into the lungs. ***Any four @1 = 04 marks***

   (iii) **Structural Support:** It provides support for the shoulder girdle and upper limbs.

   (iv) **Muscle Attachment:** It serves as an attachment site for various muscles, including the **intercostal muscles**, diaphragm, and muscles of the neck, back, and chest.

   (v) **Blood Cell Production:** The bone marrow within the ribs is a site for **haematopoiesis** (production of red and white blood cells).

Vertebral Column:

(i) Protects the spinal cord
(ii) supports the head *03 marks*
(iii)provides an attachment point for ribs and muscles.

(b) Difference: **Tendons** join muscle to bone; **Ligaments** join bone to bone at joints. *02 marks*

(Total: 09 marks)

SECTION C (30 MARKS)

9. Human Reproductive System

Introduction: *02 marks*

(a) **Hormones:**

- Male
  Testosterone - sperm production/secondary traits) *01½ marks*
- Female
  Estrogen - repair of uterine lining;
  Progesterone - maintains pregnancy/lining). *03 marks*

(b) **Stages:**

- Menstruation (shedding)
- Follicular phase (egg develops)
- Ovulation (egg release), *04 marks*
- Luteal phase (lining thickens).

(c) **Family Planning Importance:**

- Reduces maternal mortality.
- Improves economic stability for families. *03 marks*
- Helps in population control relative to national resources.

Conclusion: *01½ marks*

Total: 15 marks

10. Variation is the possession of characteristics which are different from the parent and other offspring. *02 marks*

Causes of genetic variation:

- Meiosis: during meiosis there is crossing over and segregation of different gametes. This reduces the chance of individuals being the same. .....*03 marks*
- Fertilization: during fertilization the nuclei of male and female gametes fuse. This permits parental genes to be brought together in different combinations. .....*03 marks*
- Mutation- This is a sudden change in gene which can be inherited. The individual who has undergone mutation is called a mutant.....*03 marks*

Sometimes the environmental conditions may become so adverse that the survival of most of the members of the species become difficult except for a few variants that have developed resistance. The survival of these few variants helps in the survival of the species even though so many individual members of the species have perished. *04 marks*

Total: 15 marks

11. Waste Management

Introduction: *01½ mark*

Types:

(i) Solid wastes: These are wastes which are about 75% solid. Example of solid wastes include house hold solid wastes such as vegetable, fruit and garden wastes, ….

(ii) Liquid wastes: These are wastes which are in the form of liquid or watery materials. They contain less than 25% solid substances. Example of liquid wastes include water from sinks, wash basins and baths, and urine. *04 marks*

(iii) Gaseous wastes: These are waste materials which are released into the atmosphere in gaseous state. Examples of gaseous wastes are carbon dioxide, carbon monoxide, sulphur dioxide and nitrogen dioxide.

(iv) Sludge: these are semi-solid wastes i.e. waste which is neither solids nor liquids, for example, a mixture of livestock urine and cow dung

Environmental and Health Problems of Poor Waste Disposal

(i) **Disease Outbreaks:** Unmanaged waste serves as a breeding ground for vectors such as mosquitoes, flies, and rodents. This directly increases the transmission of diseases like malaria, cholera, typhoid, and dysentery… *@1½ = 09marks*

(ii) **Water Contamination:** Haphazard dumping into rivers, streams, and drainage systems leads to polluted water sources. Toxic chemicals from hazardous or industrial waste can leach into the soil and groundwater, making it unsafe for drinking and agriculture.

(iii) **Urban Flooding:** Uncollected garbage frequently clogs drainage channels and seasonal streams. During rainy seasons, this leads to severe flooding, which spreads contaminated water throughout residential areas.

(iv) **Air Pollution:** In areas with poor collection services, residents often resort to **open burning** of trash. This releases dangerous chemicals like dioxins and furans, causing respiratory issues and long-term health complications for nearby communities.

(v) **Greenhouse Gas Emissions:** Decomposing organic matter in open, unmanaged dumpsites releases significant amounts of **methane**, a potent greenhouse gas that contributes to global warming and climate change.

(vi) **Aesthetic Degradation and Bad Odors:** Piles of uncollected waste along streets and in market places create unpleasant sights and foul smells. This degrades the quality of life for residents and can negatively impact the hospitality and tourism industries.

Conclusion: *01 mark*

Total: 15 marks